

Wednesday, May 1

Generalized Additive Models

Consider models of the form

$$g[E(Y)] = \beta_0 + f(x)$$

or

$$g[E(Y)] = \beta_0 + f_1(x_1) + f_2(x_2)$$

or

$$g[E(Y)] = \beta_0 + f(x_1, x_2)$$

or

$$g[E(Y)] = \beta_0 + f_1(x_1) + f_2(x_2, x_3),$$

where g is a link function and f , f_1 , and f_2 are functions. Linear and generalized linear models are special cases of GAMs. But the term GAM usually refers to cases where the functions of the explanatory variables are specified to be flexible but smooth functions. Splines are frequently used for these functions.

Splines

A spline can be viewed a couple of different ways.

1. A function made up of several polynomial functions that join at a set of knots.

Example: A *cubic spline* for a linear model with knots ζ_1 and ζ_2 can be written as

$$E(Y) = \begin{cases} \delta_0 + \delta_1 x + \delta_2 x^2 + \delta_3 x^3, & \text{if } x < \zeta_1, \\ \delta_0 + \delta_1 x + \delta_2 x^2 + \delta_3 x^3 + \delta_4 (x - \zeta_1)^3, & \text{if } \zeta_1 \leq x < \zeta_2 \\ \delta_0 + \delta_1 x + \delta_2 x^2 + \delta_3 x^3 + \delta_4 (x - \zeta_1)^3 + \delta_5 (x - \zeta_2)^3, & \text{if } \zeta_2 \leq x. \end{cases}$$

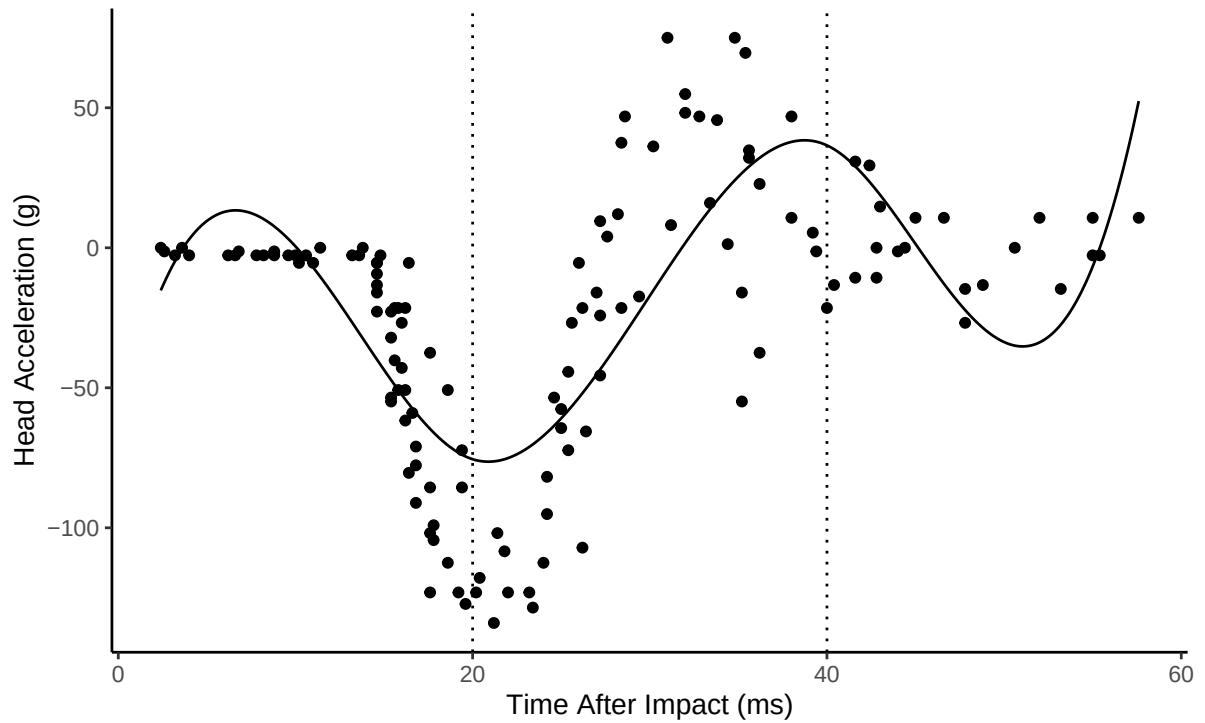
Here is cubic spline as a regression model.

```
library(MASS) # for the mcycle data
library(splines) # for the bs function
m <- lm(accel ~ bs(times, knots = c(20,40)), data = mcycle)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-15.22	16.36	-0.9302	3.541e-01
bs(times, knots = c(20, 40))1	86.03	28.38	3.0318	2.947e-03
bs(times, knots = c(20, 40))2	-201.34	20.74	-9.7086	5.256e-17
bs(times, knots = c(20, 40))3	198.96	29.52	6.7404	4.966e-10
bs(times, knots = c(20, 40))4	-110.05	27.77	-3.9630	1.228e-04
bs(times, knots = c(20, 40))5	67.61	29.55	2.2880	2.379e-02

```
d <- data.frame(times = seq(2.4, 57.6, length = 1000))
d$yhat <- predict(m, newdata = d)
p <- ggplot(mcycle, aes(x = times, y = accel)) +
  geom_point() + theme_classic() +
  labs(x = "Time After Impact (ms)", y = "Head Acceleration (g)") +
  geom_vline(xintercept = c(20,40), linetype = 3) +
```

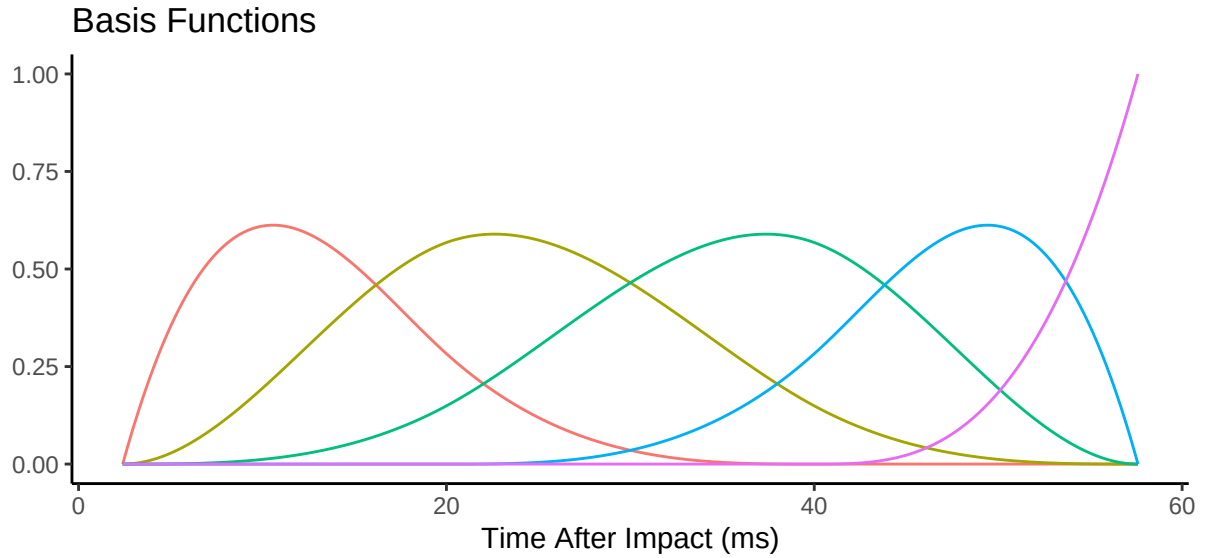
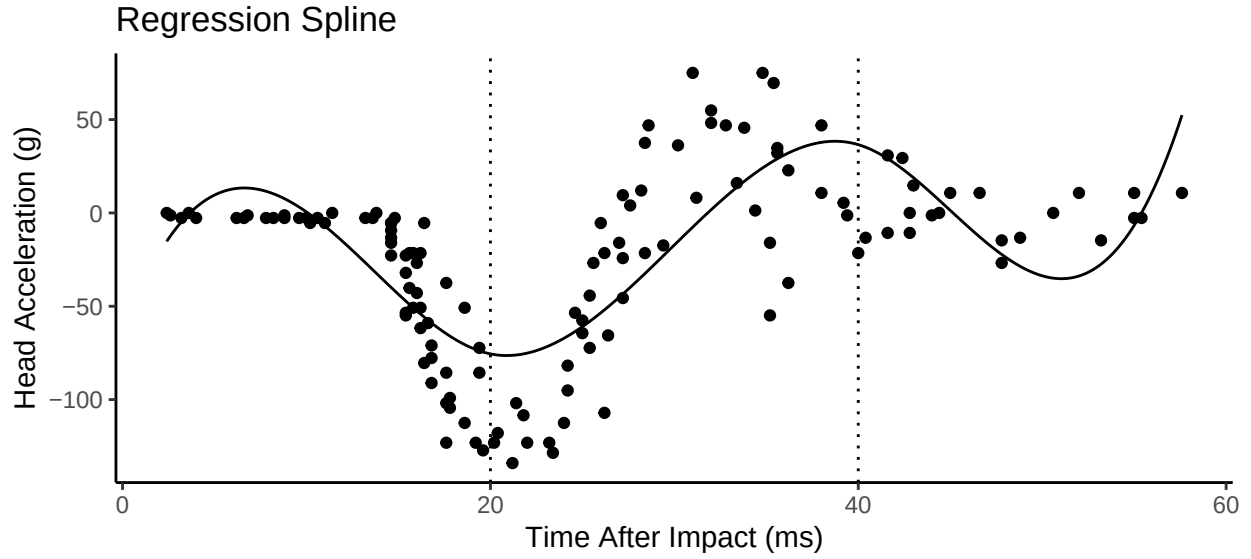
```
geom_line(aes(y = yhat), data = d)
plot(p)
```



2. A function that is a weighted average of a set of basis functions such that

$$f(x) = \sum_j \delta_j b_j(x),$$

where $b_j(x)$ is the j -th basis function and δ_j is a parameter. The spline shown above can be written in terms of five basis functions.



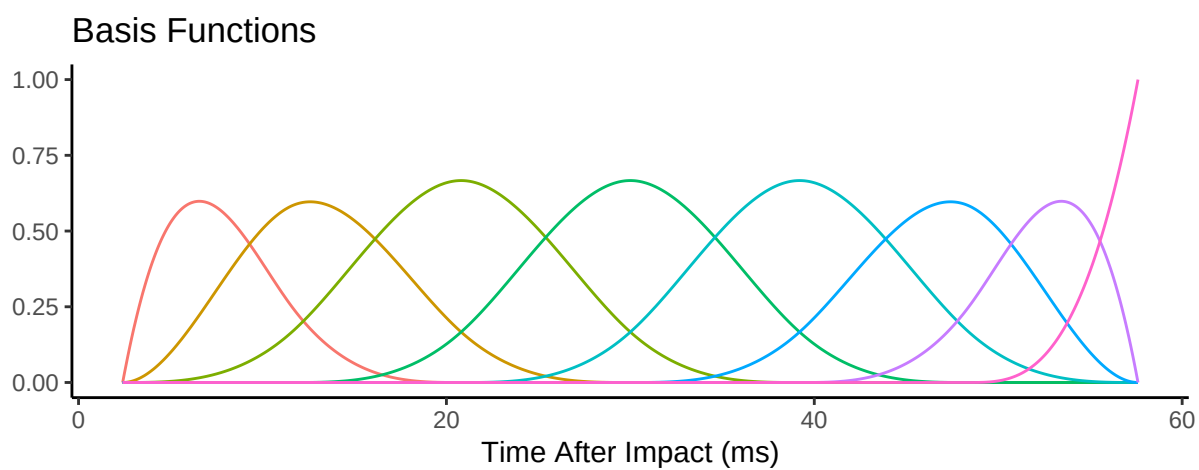
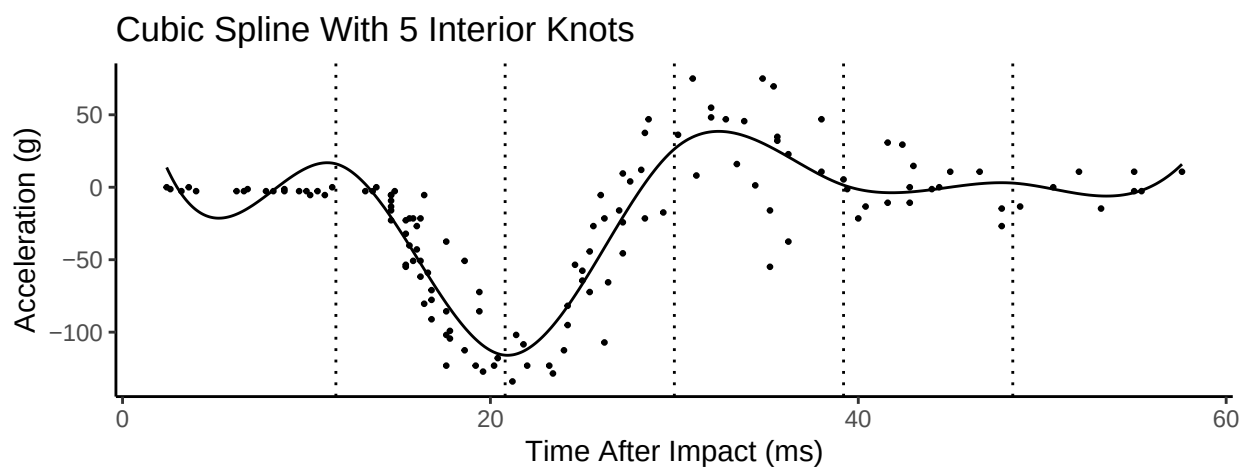
So the model can be written as

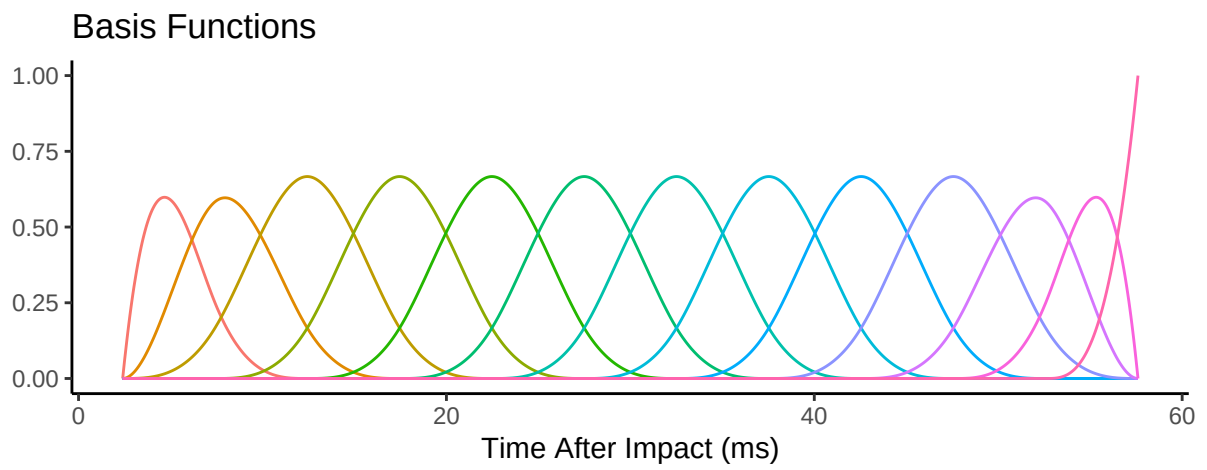
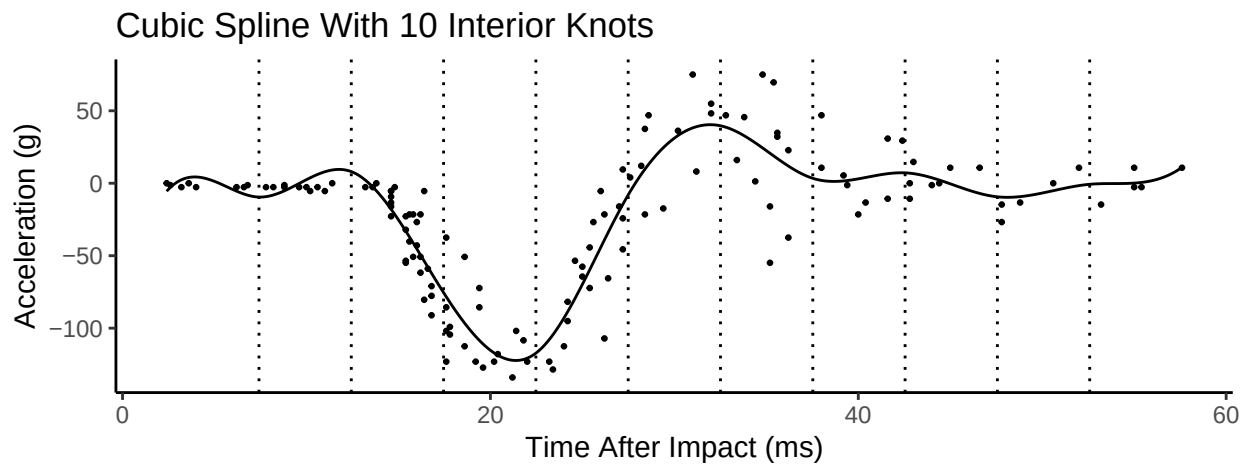
$$E(Y) = \beta_0 + \sum_{j=1}^5 \delta_j b_j(x) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \beta_3 x_3^* + \beta_4 x_4^* + \beta_5 x_5^*,$$

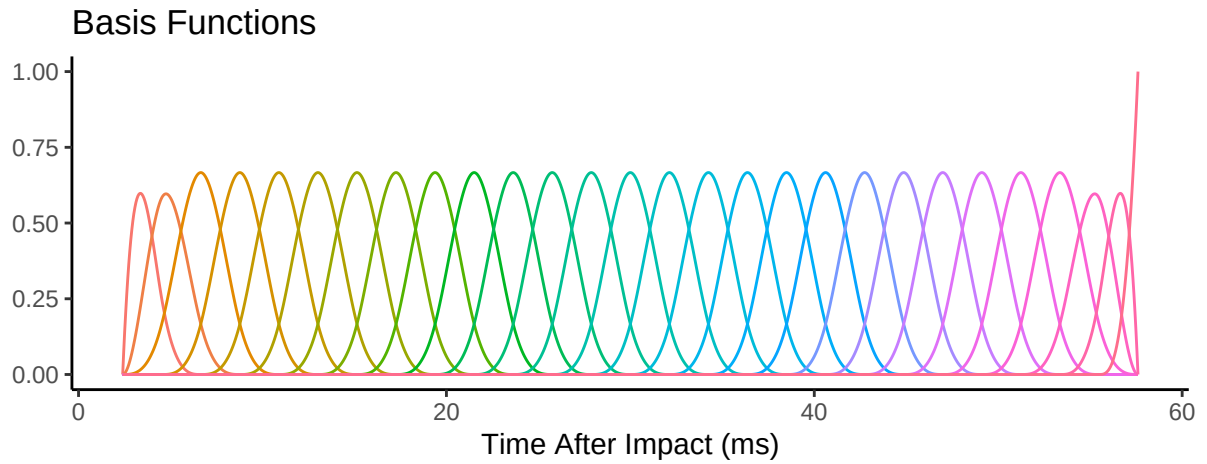
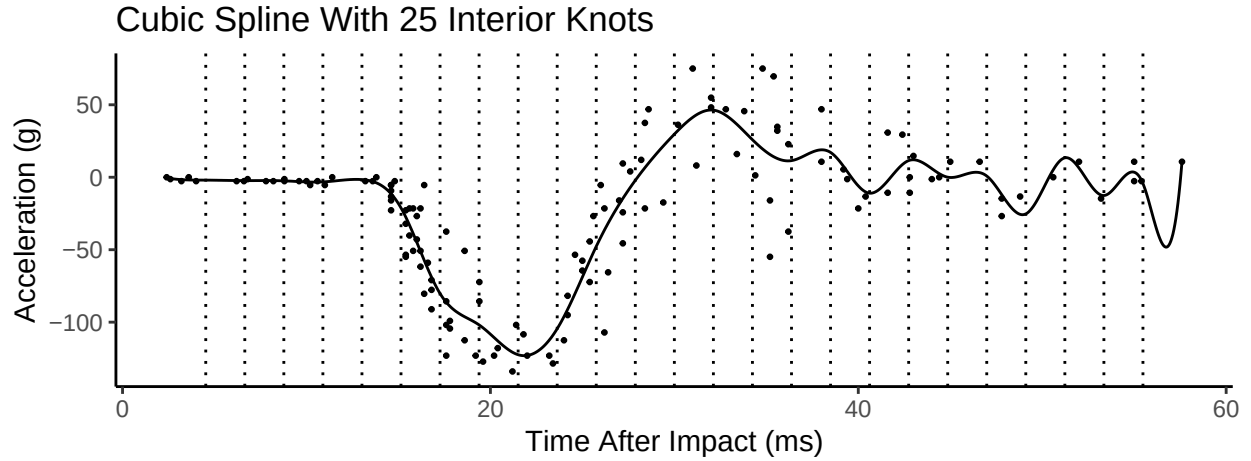
where $\delta_j = \beta_j$ and $x_j^* = b_j(x)$. Because this is still a (generalized) linear model, it is still quite tractable computationally and theoretically (provided we treat the number and placement of knots as well as the form of the functions as *known*).

Spline Complexity

The spline can be made more flexible by adding more knots or basis functions. Adding more knots or basis functions makes the spline more flexible, but potentially *too flexible*. This is a bias-variance trade-off.







In principle we could use cross-validation or a related technique (e.g. AIC) to try to identify a good trade-off. But a better approach is to use *penalization/regularization*.

Penalized Splines

Instead of trying to select the number of knots or basis functions, we could specify a “generous” number of knots/functions and introduce a penalty for “wiggleness” in the estimation. Suppose we have the model

$$E(Y_i) = \beta_0 + f(x_i),$$

where f is a spline such that

$$f(x_i) = \sum_j \delta_j b_j(x_i).$$

Then using (weighted) least squares we can try to minimize

$$\sum_{i=1}^n w_i (y_i - \hat{y}_i)^2 + \lambda h(f).$$

where $\lambda \geq 0$ and h is a function that measures the “wiggleness” of the function f . One reasonable measure of “wiggleness” is to integrate over the second derivative of f such that

$$h(f) = \int f''(x)^2 dx.$$

Fortunately this function can be written in a relatively simple way so that it is relatively easy to compute and solve the (penalized) least squares problem. The control over the flexibility/wiggleness of the spline is

then through λ . As λ increases f approaches a line, but as λ approaches zero then f becomes increasingly flexible/wiggly.

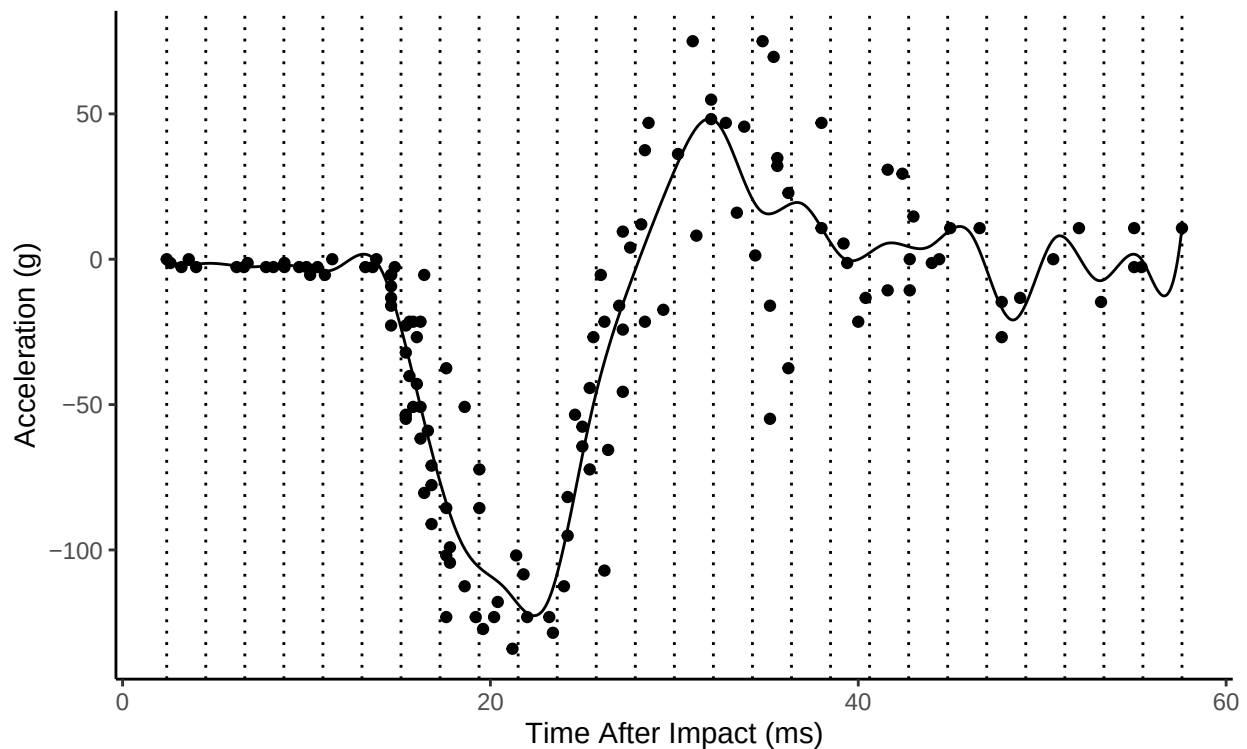
Example: Using the `gam` function from the `mgcv` package allows us to control the wiggleness through the `sp` argument. Here is “maximum wiggleness”.

```
library(mgcv) # for the gam and supporting functions

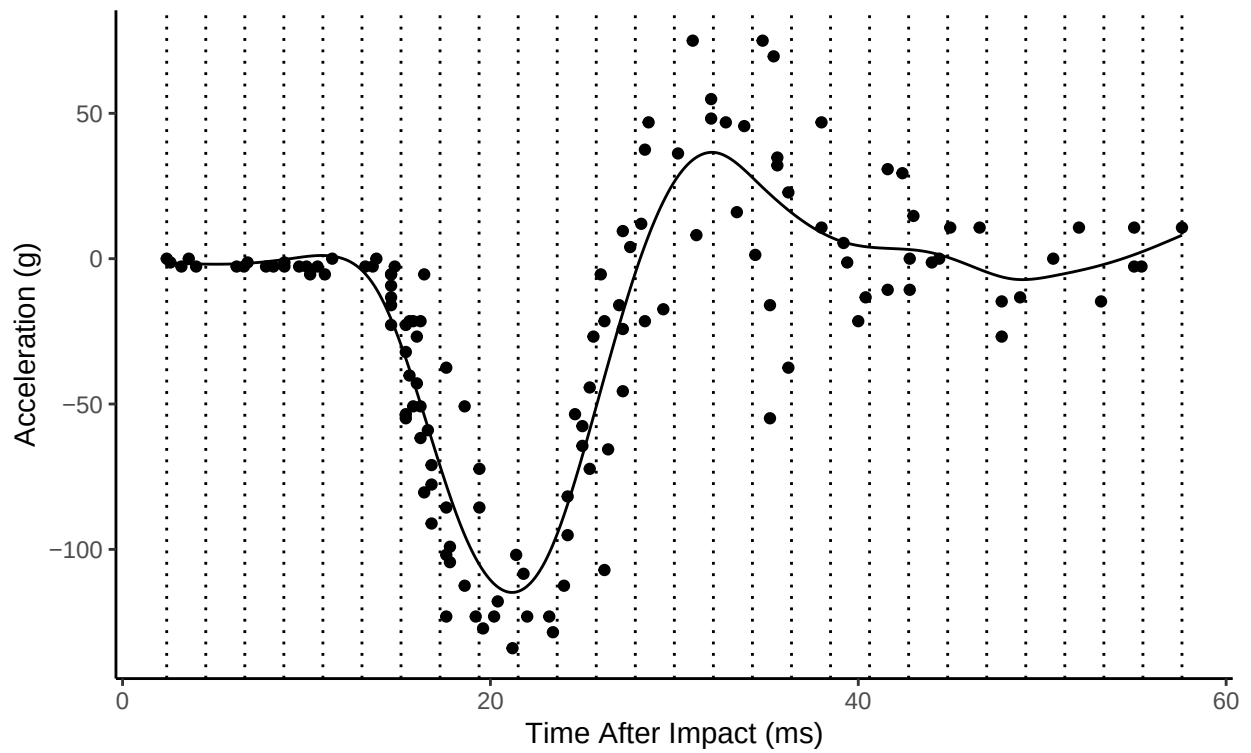
knots <- seq(2.4, 57.6, length = 27)
m <- gam(accel ~ s(times, bs = "bs", k = length(knots), sp = 0),
  data = mcycle, knots = list(x = knots))

d <- data.frame(times = seq(2.4, 57.6, length = 1000))
d$yhat <- predict(m, newdata = d)

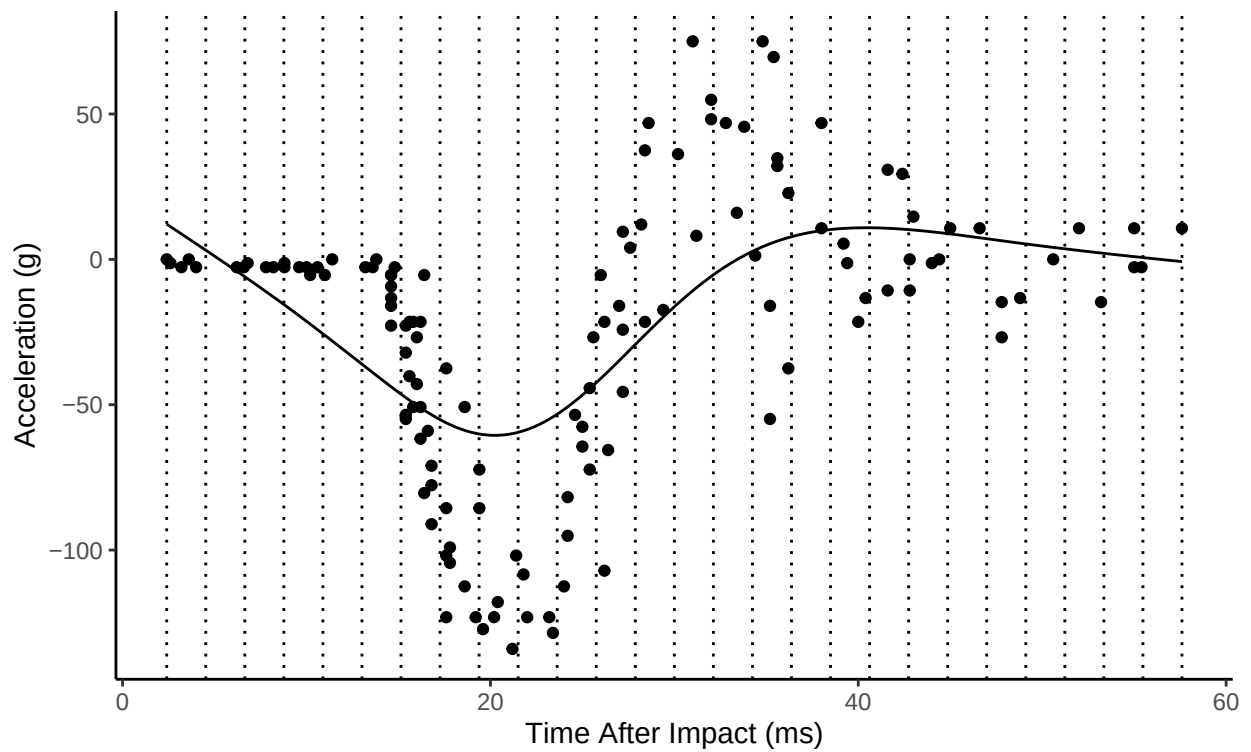
p <- ggplot(mcycle, aes(x = times, y = accel)) + theme_classic() +
  geom_point() + labs(x = "Time After Impact (ms)", y = "Acceleration (g)") +
  geom_line(aes(y = yhat), data = d) + geom_vline(xintercept = knots, linetype = 3)
plot(p)
```



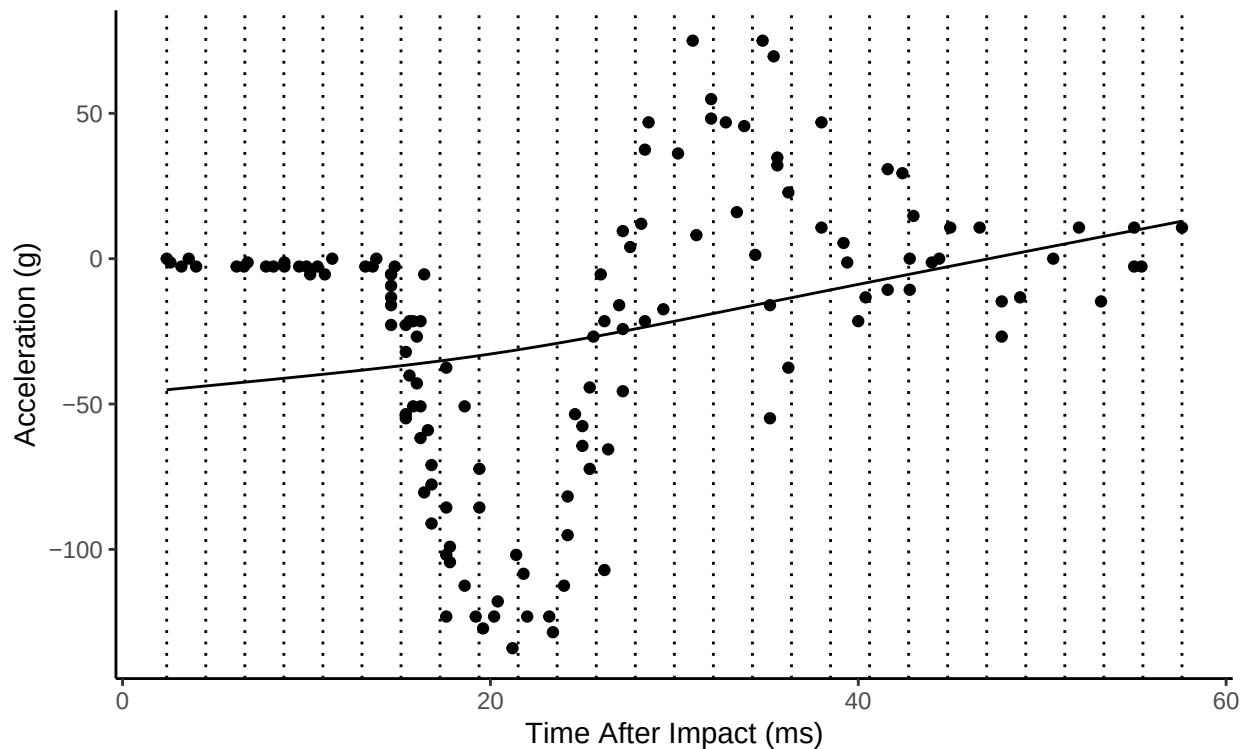
Here is the estimated model with `sp = 10`.



Here is the estimated model with $sp = 1000$.



Here is the estimated model with $sp = 100000$ (nearly minimum wiggleness).



The **mgcv** package gives the user access to a wide variety of types of splines and ways to modify them. But it also provides “automatic” cross-validation and selection of λ using a *generalized* cross-validation (GCV) measure.

Example: Consider again the `mcycles` data. Here we will use the default settings.

```
m <- gam(accel ~ s(times), data = mcycle)
summary(m)
```

Family: gaussian

Link function: identity

Formula:

accel ~ s(times)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-25.55	1.95	-13.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(times)	8.69	8.97	53.5	<2e-16 ***

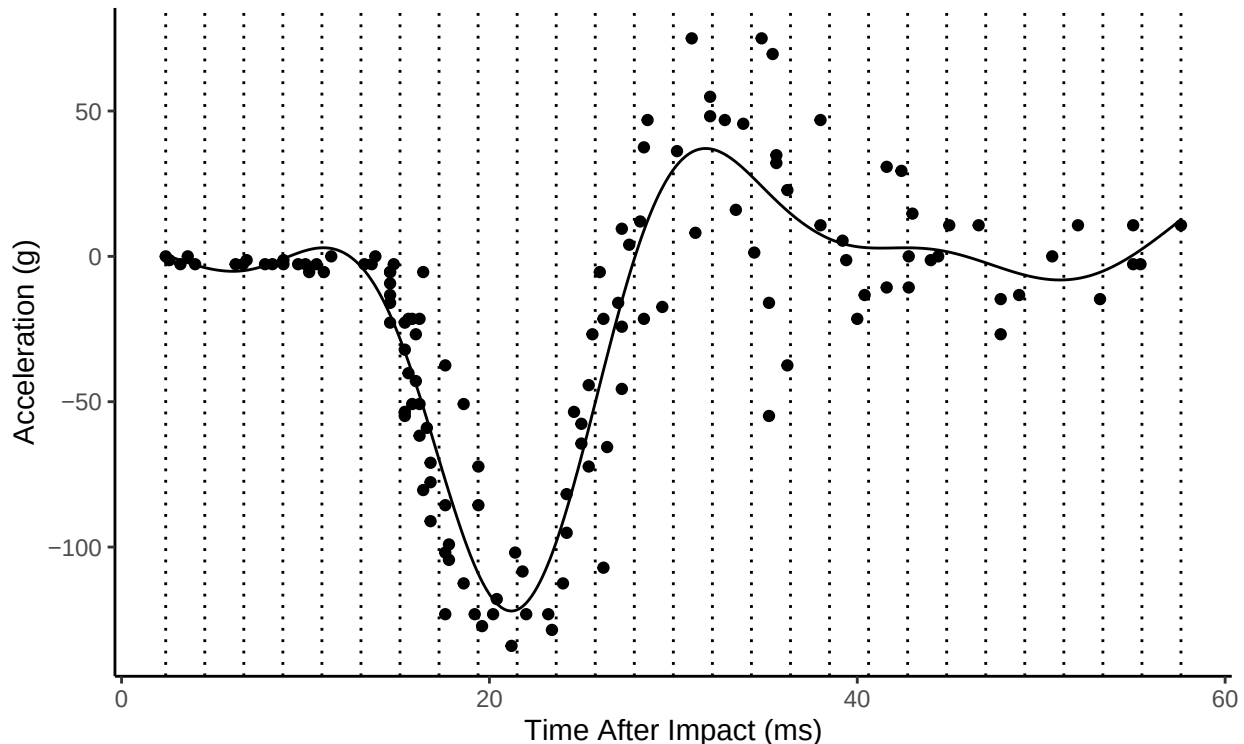
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.783 Deviance explained = 79.8%

GCV = 545.78 Scale est. = 506 n = 133

```
d <- data.frame(times = seq(2.4, 57.6, length = 1000))
d$yhat <- predict(m, newdata = d)
```

```
p <- ggplot(mcycle, aes(x = times, y = accel)) + theme_classic() +
  geom_point() + labs(x = "Time After Impact (ms)", y = "Acceleration (g)") +
  geom_line(aes(y = yhat), data = d) +
  geom_vline(xintercept = knots, linetype = 3)
plot(p)
```



Example: Consider the bone data that shows relative bone mineral density by age and sex.

```
bone <- read.table("http://faculty.washington.edu/jonno/book/spinalbonedata.txt", header = TRUE)
m <- gam(spnbmd ~ s(age), data = bone)
```

```
d <- expand.grid(sex = c("fem", "mal"), age = seq(9, 26, length = 1000))
d$yhat <- predict(m, newdata = d)
```

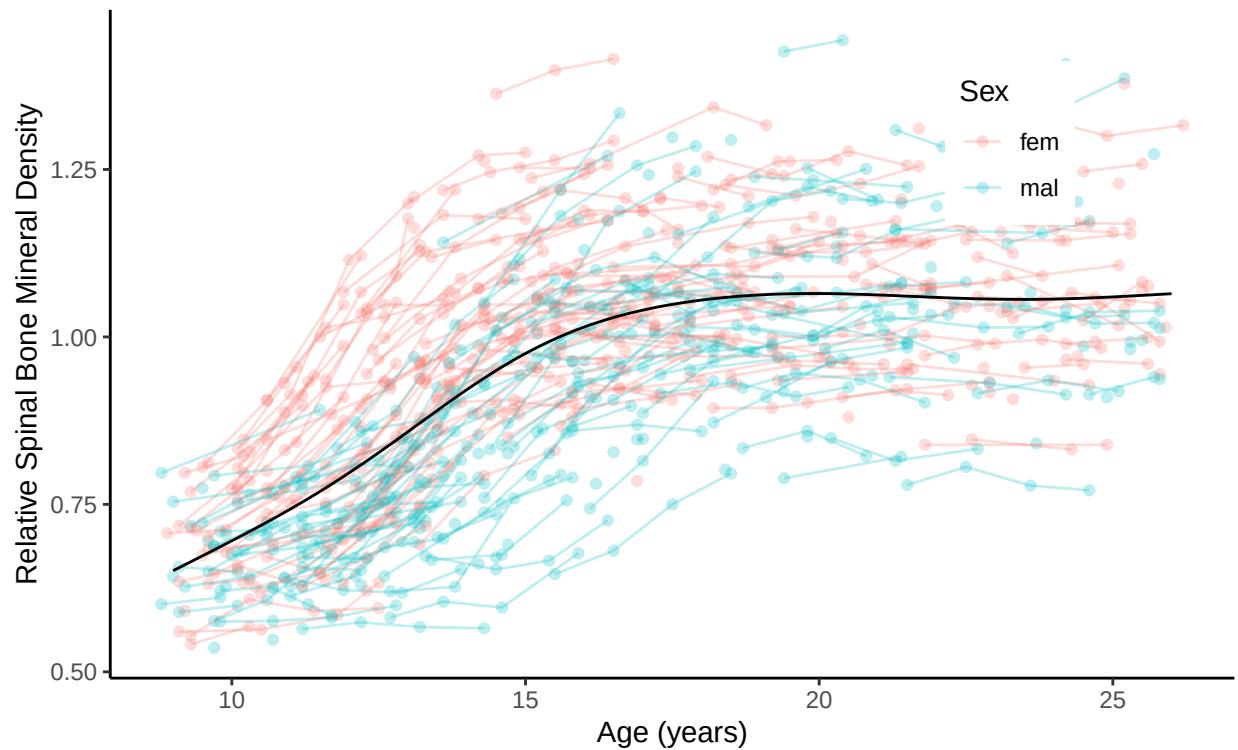
```
p <- ggplot(bone, aes(x = age, y = spnbmd)) +
  geom_point(aes(color = sex), alpha = 0.25) + theme_classic() +
  geom_line(aes(color = sex, group = idnum), alpha = 0.25) +
  theme(legend.position = c(0.8, 0.8)) +
  labs(x = "Age (years)", color = "Sex",
       y = "Relative Spinal Bone Mineral Density") +
  geom_line(aes(y = yhat), data = d)
```

Warning: A numeric `legend.position` argument in `theme()` was deprecated in ggplot2 3.5.0.
Please use the `legend.position.inside` argument of `theme()` instead.

This warning is displayed once every 8 hours.

Call `lifecycle::last\_lifecycle\_warnings()` to see where this warning was generated.

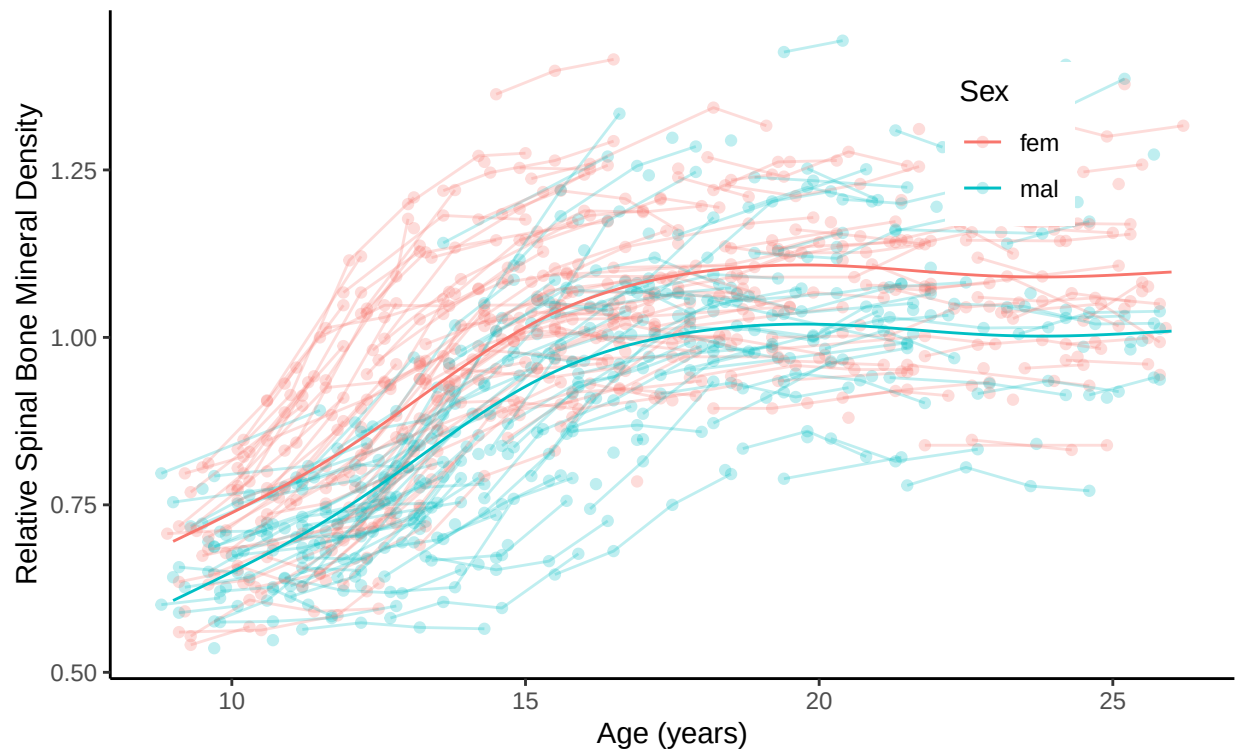
```
plot(p)
```



```
m <- gam(spnbmd ~ sex + s(age), data = bone)

d <- expand.grid(sex = c("fem", "mal"), age = seq(9, 26, length = 1000))
d$yhat <- predict(m, newdata = d)

p <- ggplot(bone, aes(x = age, y = spnbmd)) +
  geom_point(aes(color = sex), alpha = 0.25) + theme_classic() +
  geom_line(aes(color = sex, group = idnum), alpha = 0.25) +
  theme(legend.position = c(0.8, 0.8)) +
  labs(x = "Age (years)", color = "Sex",
       y = "Relative Spinal Bone Mineral Density") +
  geom_line(aes(y = yhat, color = sex), data = d)
plot(p)
```



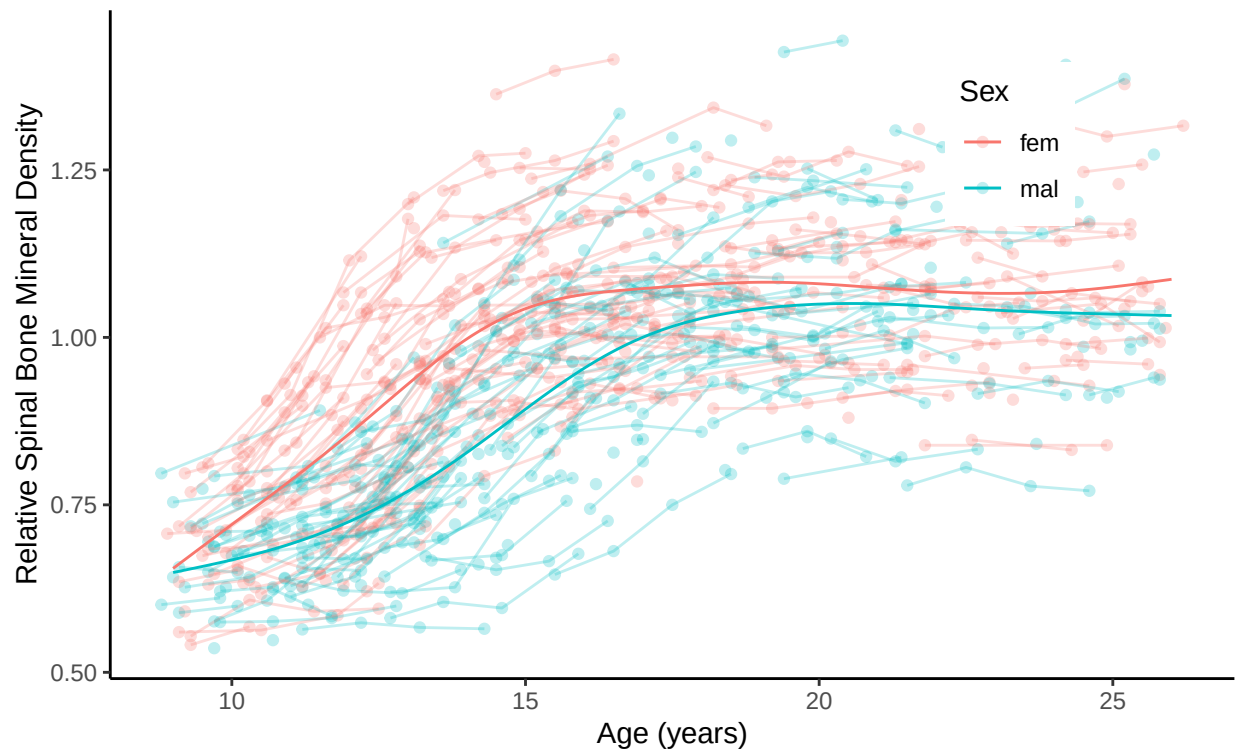
```
m <- gam(spnbmd ~ sex + s(age, by = factor(sex)), data = bone)

d <- expand.grid(sex = c("fem", "mal"), age = seq(9, 26, length = 1000))
d$yhat <- predict(m, newdata = d)

p <- ggplot(bone, aes(x = age, y = spnbmd)) +
  geom_point(aes(color = sex), alpha = 0.25) + theme_classic() +
  geom_line(aes(color = sex, group = idnum), size = 0.5, alpha = 0.25) +
  theme(legend.position = c(0.8, 0.8)) +
  labs(x = "Age (years)", color = "Sex",
       y = "Relative Spinal Bone Mineral Density") +
  geom_line(aes(y = yhat, color = sex), data = d)
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.
 This warning is displayed once every 8 hours.
 Call `lifecycle::last\_lifecycle\_warnings()` to see where this warning was generated.

```
plot(p)
```

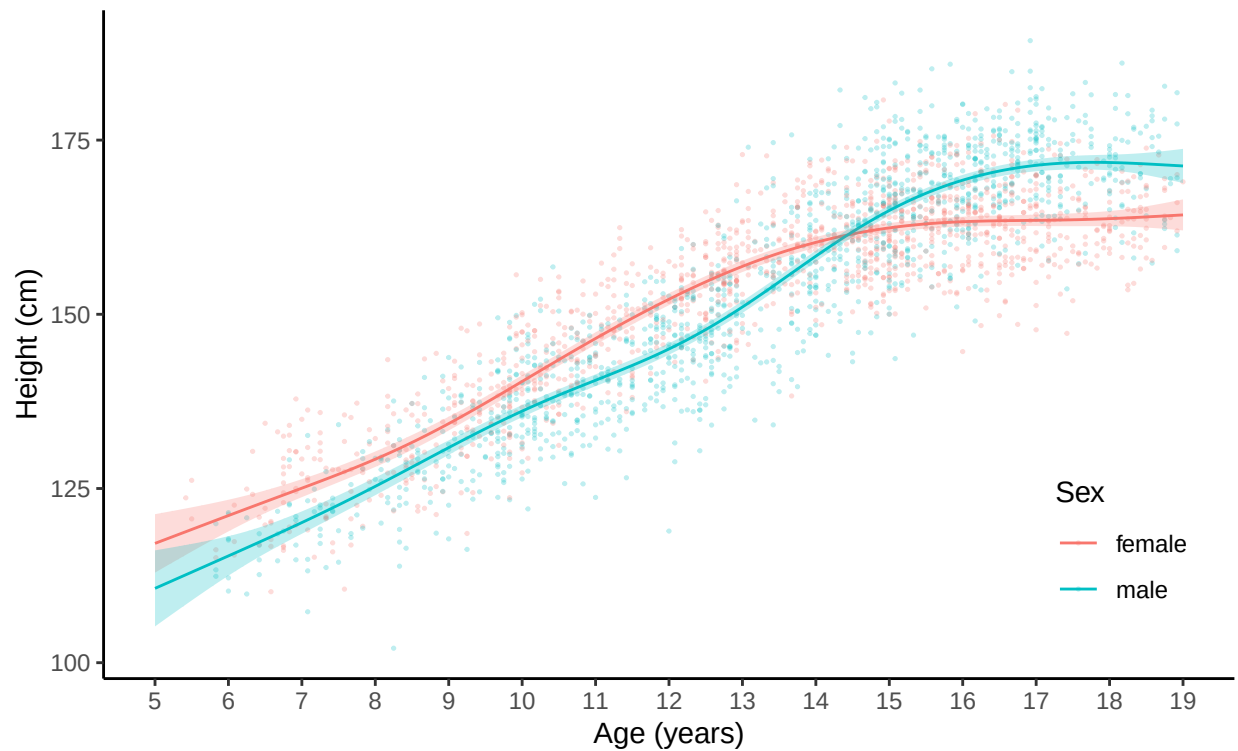


Example: Consider the growth data of female and male children in the `children` data frame from the `npregfast` package. Here I am adding a confidence band to each function.

```
library(npregfast)
m <- gam(height ~ sex + s(age, by = factor(sex)), data = children)

d <- expand.grid(age = seq(5, 19, length = 1000),
  sex = c("female", "male"))
d$yhat <- predict(m, newdata = d)
d$se <- predict(m, newdata = d, se.fit = TRUE)$se.fit
d$lower <- d$yhat - 2*d$se
d$upper <- d$yhat + 2*d$se

p <- ggplot(children, aes(x = age, y = height)) + theme_classic() +
  geom_point(aes(color = sex), size = 0.25, alpha = 0.25) +
  geom_line(aes(y = yhat, color = sex), data = d) +
  geom_ribbon(aes(x = age, ymin = lower, ymax = upper,
    fill = sex, y = NULL), data = d, color = NA, alpha = 0.25) +
  labs(x = "Age (years)", y = "Height (cm)", color = "Sex") +
  theme(legend.position = c(0.9, 0.2)) +
  scale_x_continuous(breaks = seq(5, 19, by = 1)) + guides(fill = "none")
plot(p)
```



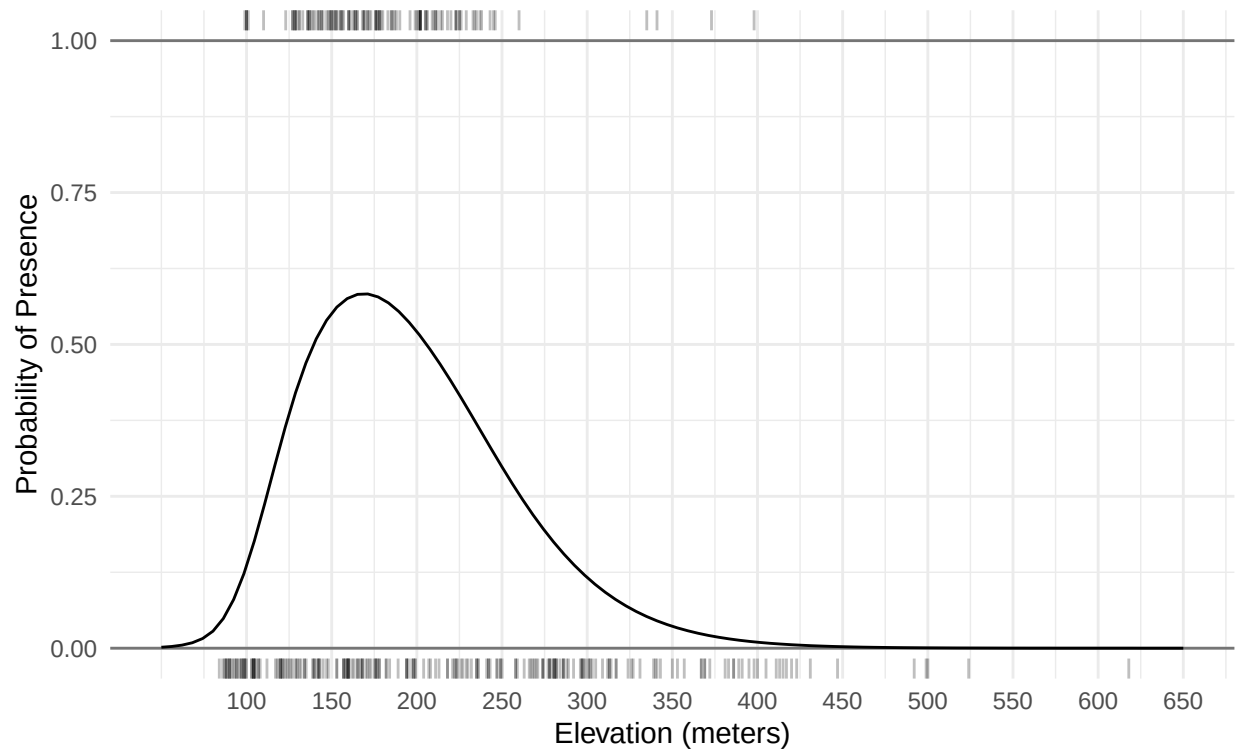
The **scam** package can be used to estimate shape-constrained generalized additive models (e.g., monotonic and/or concave or convex).

```
library(scam)
library(blmecco)
data(anoctua)

m <- scam(PA ~ s(elevation, k = 12, bs = "cv", m = 2), family = binomial, data = anoctua)

d <- data.frame(elevation = seq(50, 650, length = 100))
d$yhat <- predict(m, newdata = d, type = "response")

p <- ggplot(anoctua, aes(x = elevation, y = PA)) + theme_minimal() +
  geom_rug(data = subset(anoctua, PA == 0), alpha = 0.25, sides = "b") +
  geom_rug(data = subset(anoctua, PA == 1), alpha = 0.25, sides = "t") +
  geom_hline(yintercept = c(0, 1), alpha = 0.5) +
  labs(x = "Elevation (meters)", y = "Probability of Presence") +
  scale_x_continuous(breaks = seq(100, 700, by = 50)) +
  geom_line(aes(y = yhat), data = d)
plot(p)
```



Example: Multiple explanatory variables can be “smoothed” in a GAM. For example, consider the model

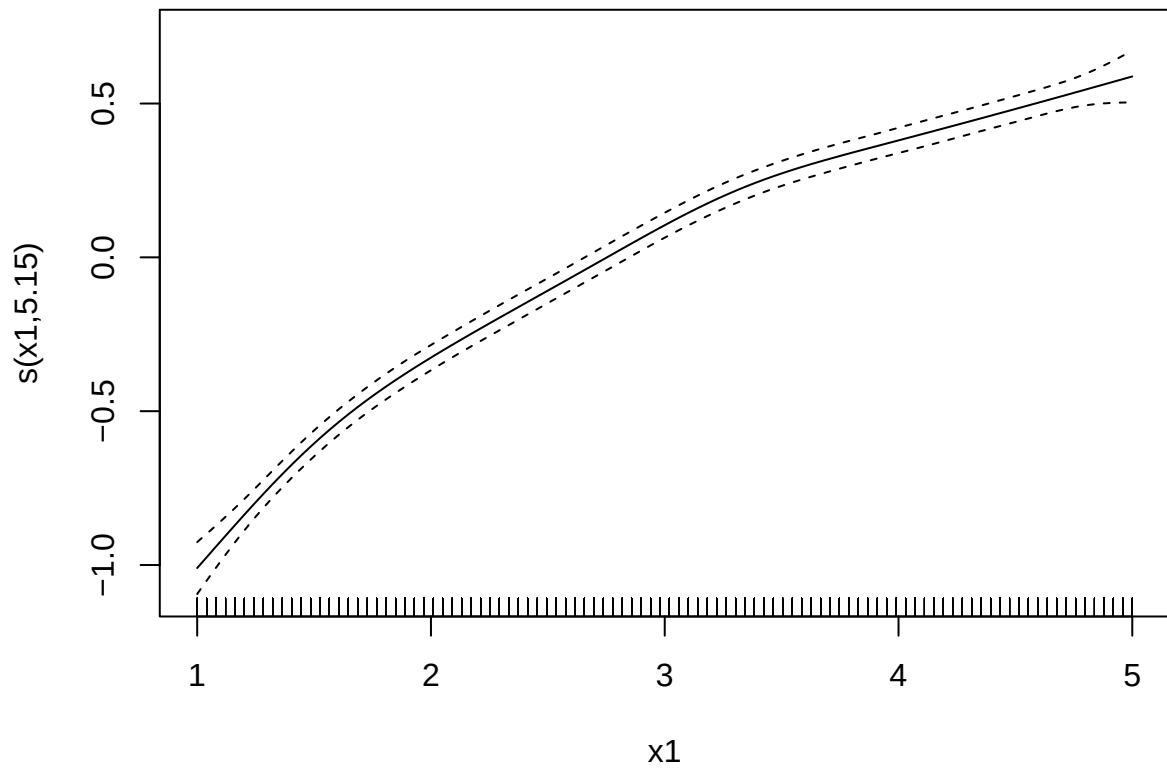
$$E(Y) = \beta_0 + f_1(x_1) + f_2(x_2),$$

where $\beta_0 = 5$, $f_1(x) = \log(x_1)$, and $f_2(x_2) = 0.05x_2^2$. But suppose we don’t know the functions f_1 and f_2 but instead *estimate* them from the data.

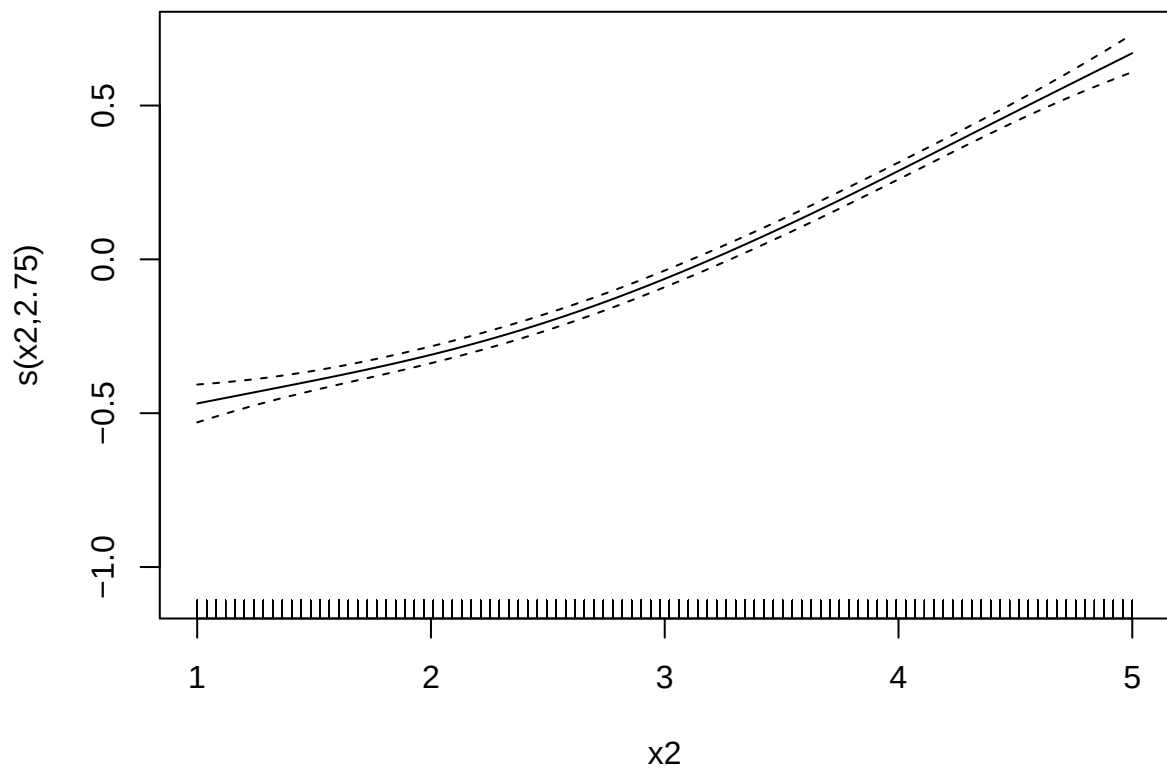
```
set.seed(123)

d <- expand.grid(x1 = seq(1, 5, length = 100), x2 = seq(1, 5, length = 100))
d$y <- with(d, 5 + log(x1) + 0.05 * x2^2 + rnorm(nrow(d)))

m <- gam(y ~ s(x1) + s(x2), data = d)
plot(m, select = 1)
```



```
plot(m, select = 2)
```



Example: Let's revisit the crab experiment.

```
library(Stat2Data)
data(CrabShip)
```

```
m <- gam(Oxygen ~ Noise + s(Mass, by = Noise), data = CrabShip)
summary(m)
```

Family: gaussian
Link function: identity

Formula:
Oxygen ~ Noise + s(Mass, by = Noise)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	166.68	8.06	20.67	< 2e-16 ***
Noiseship	74.82	11.47	6.52	3.8e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

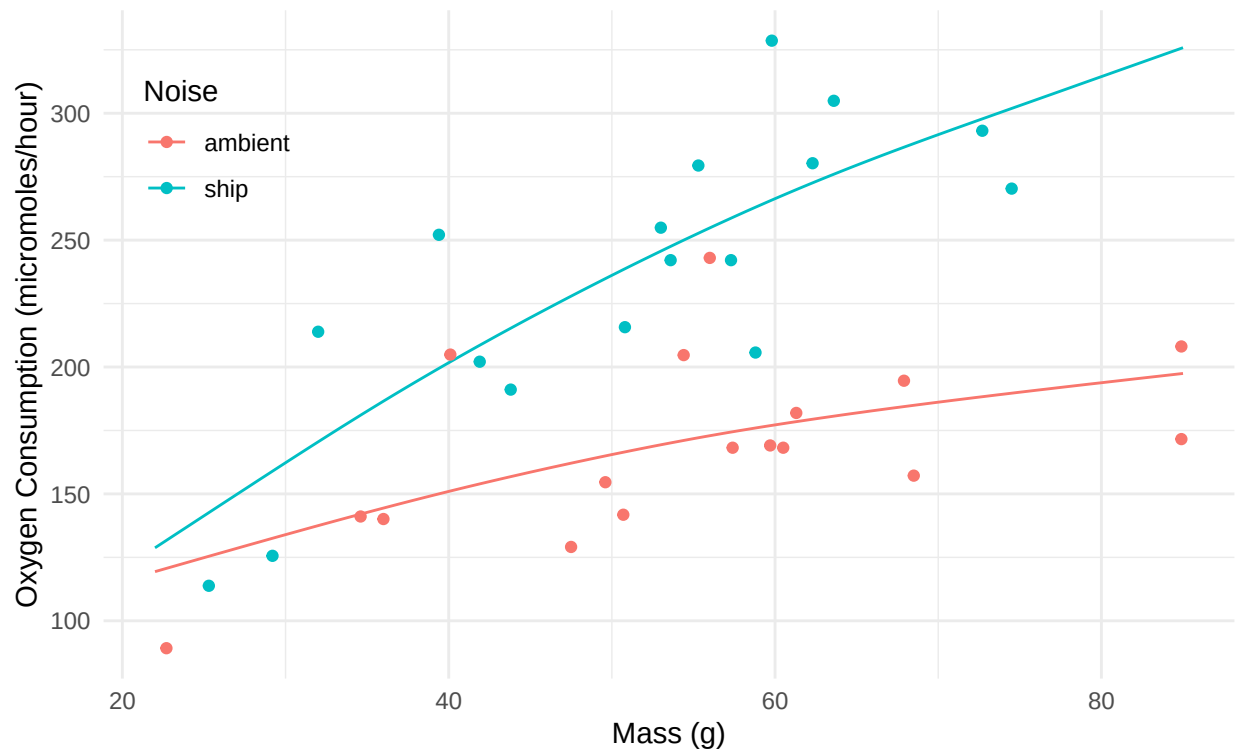
	edf	Ref.df	F	p-value
s(Mass):Noiseambient	1.45	1.76	4.18	0.054 .
s(Mass):Noiseship	1.54	1.89	15.89	2.4e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.685 Deviance explained = 72.3%
GCV = 1276 Scale est. = 1088.8 n = 34

```
d <- expand.grid(Mass = seq(22, 85, length = 100), Noise = c("ambient", "ship"))
d$yhat <- predict(m, newdata = d)
```

```
p <- ggplot(CrabShip, aes(x = Mass, y = Oxygen, color = Noise)) +
  geom_line(aes(y = yhat), data = d) +
  geom_point() + theme_minimal() +
  labs(y = "Oxygen Consumption (micromoles/hour)", x = "Mass (g)") +
  theme(legend.position = c(0.1, 0.8))
plot(p)
```



```
library(emmeans)
pairs(emmeans(m, ~Noise|Mass, at = list(Mass = c(40,50,60))),
      reverse = TRUE, infer = TRUE)
```

Mass = 40:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
ship - ambient	50.7	16.1	29	17.8	83.6	3.154	0.0037

Mass = 50:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
ship - ambient	70.6	13.5	29	43.0	98.3	5.221	<.0001

Mass = 60:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
ship - ambient	89.1	13.6	29	61.3	117.0	6.546	<.0001

Confidence level used: 0.95

Example: How about the fruit fly data (a GAM ANCOVA)?

```
library(faraway)
m <- gam(longevity ~ activity + s(thorax), data = fruitfly)
summary(m)
```

Family: gaussian

Link function: identity

Formula:

longevity ~ activity + s(thorax)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.04	2.12	29.21	< 2e-16 ***
activityone	2.01	3.01	0.67	0.505
activitylow	-7.34	2.98	-2.46	0.015 *
activitymany	4.03	3.03	1.33	0.186
activityhigh	-20.47	3.03	-6.76	5.8e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(thorax)	2.86	3.57	32.1	<2e-16 ***

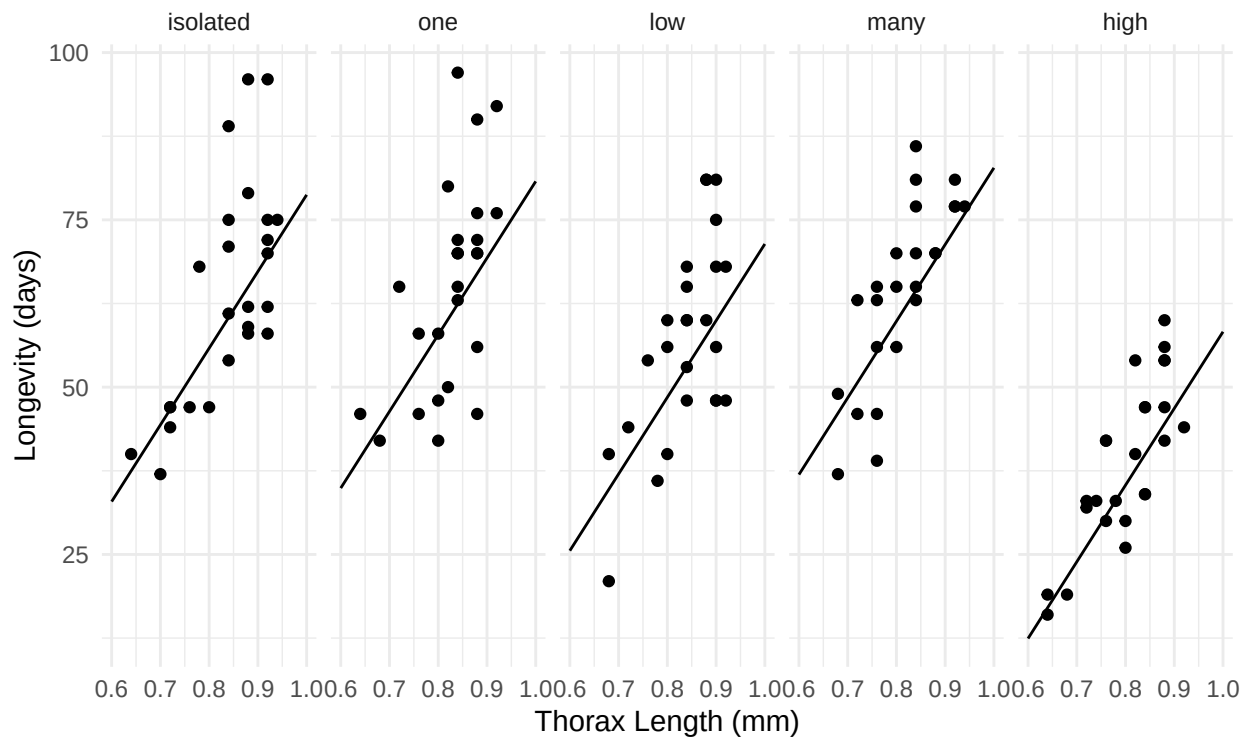
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.643 Deviance explained = 66.3%

GCV = 116.92 Scale est. = 109.51 n = 124

```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.6,1))
d$yhat <- predict(m, newdata = d, type = "response")

p <- ggplot(fruitfly, aes(x = thorax, y = longevity)) + theme_minimal() +
  geom_point() + facet_wrap(~ activity, ncol = 5) +
  labs(x = "Thorax Length (mm)", y = "Longevity (days)") +
  geom_line(aes(y = yhat), data = d)
plot(p)
```



Still fairly linear though!